The Importance of Computational Discourse Analysis

Abstract

Computational Discourse Analysis (CDA) has become an essential field of inquiry in response to the growing complexity and scale of textual data in contemporary societies. Traditional discourse analysis has offered rich qualitative insights into meaning, ideology, and social interaction, yet it faces methodological limitations when applied to large and heterogeneous corpora. This review article argues that computational discourse analysis provides a necessary extension to classical discourse studies by integrating cognitive theories of discourse comprehension, linguistically informed annotated resources, and large-scale computational methods. Drawing on three representative works—a theoretical account of discourse as resource-limited inferential processing, the construction of a discourse and coreference corpus for Persian, and a comparative computational analysis of public discourse on artificial intelligence—this article demonstrates how computational discourse analysis bridges theory, data, and social application. The review highlights the epistemological, methodological, and societal significance of CDA and outlines its implications for future research.

Keywords: computational discourse analysis, artificial intelligence, social interaction, computational methods.

1. Introduction

Discourse analysis has long occupied a central position in linguistics and the social sciences, addressing how language functions beyond isolated sentences to construct coherent meanings, social identities, and power relations.

Approaches such as pragmatics, conversation analysis, and critical discourse analysis have emphasized interpretation, context, and ideology, offering nuanced readings of texts and interactions. However, the communicative landscape has changed dramatically in recent decades. Digital platforms, social media, and automated communication systems now generate vast volumes of text that exceed the scope of traditional qualitative analysis. In this context, computational discourse analysis emerges not merely as a technical innovation but as a conceptual necessity. It seeks to operationalize discourse-theoretical concepts such as coherence,

reference, and inference—in computational terms, enabling systematic and scalable analysis.

This article reviews three influential contributions that collectively illustrate the importance of computational discourse analysis as a field that connects cognitive theory, linguistic structure, and social meaning.

2. Discourse Analysis and the Computational Turn

Early discourse analysis developed largely independently of computational linguistics. While computational models focused on syntax and semantics at the sentence level, discourse analysts emphasized interpretation, social context, and ideological critique. This separation led to a methodological divide: computational approaches were often criticized for oversimplifying meaning, while discourse-analytic approaches were limited in scale and reproducibility. The rise of natural language processing and machine learning has intensified interest in discourse-level phenomena. Yet purely data-driven models risk treating discourse as a collection of surface patterns rather than a structured meaning-making process. Computational discourse analysis responds to this tension by insisting that discourse must be modeled as both a cognitive and social phenomenon. The three works examined in this review exemplify how this integration can be achieved.

3. Theoretical Foundations: Discourse as Inferential Processing

A key contribution to the theoretical grounding of computational discourse analysis is the reconceptualization of discourse comprehension as a form of inferential reasoning. Webber's framework characterizes discourse understanding as a resource-limited inferential search, in which readers or listeners generate possible interpretations and evaluate them under cognitive constraints. This perspective has profound implications for computational modeling. It suggests that discourse relations, reference resolution, and coherence are not simply encoded in linguistic forms but are actively constructed through inference. As a result, effective computational models must approximate this inferential process, even when implemented through probabilistic or neural mechanisms. Rather than viewing discourse markers or coherence relations as static labels, this approach treats them as cues that guide interpretation. By grounding computational discourse analysis in cognitive theory, this framework prevents the field from becoming purely technical.

It emphasizes that computational models are not ends in themselves, but tools for approximating human meaning-making processes.

4. Linguistic Resources: Discourse and Coreference Annotation

While theoretical models provide conceptual clarity, computational discourse analysis depends critically on the availability of high-quality annotated data. The construction of discourse treebanks represents a major methodological advance, allowing abstract discourse relations to be encoded in machine-readable form. The Persian Discourse Treebank and Coreference Corpus illustrates the importance of such resources, particularly for under-resourced languages. By adapting the Penn Discourse Treebank framework to Persian, this work demonstrates that discourse relations such as causality, contrast, and elaboration can be systematically annotated across languages, despite typological differences. Equally important is the inclusion of document-level coreference annotation. Tracking entities across extended texts is central to discourse coherence, yet it poses significant challenges for computational models—especially in languages like Persian, which allow subject omission and lack grammatical gender. This corpus not only enables the training of computational models but also advances theoretical understanding of discourse structure in Persian. More broadly, this contribution underscores that computational discourse analysis is inherently linguistic. Successful modeling requires sensitivity to language-specific features, rather than reliance on universal or language-agnostic assumptions.

5. Applied Computational Discourse Analysis: Social Meaning at Scale

The societal relevance of computational discourse analysis becomes most apparent in applied studies that examine real-world discourses. The comparative analysis of hopes and fears surrounding artificial intelligence exemplifies how computational methods can reveal patterns that are inaccessible to manual analysis. By combining topic modeling with sentiment analysis, this study captures both the thematic structure and the emotional orientation of AI-related discourse across different social actors. Importantly, it shows that discourse is not uniform: political institutions, corporate actors, and online communities construct distinct narratives, each reflecting specific interests and values. This work demonstrates that computational discourse analysis can extend the goals of critical discourse analysis by providing empirical grounding at scale. Rather than replacing interpretive analysis, computational methods offer a complementary perspective that identifies patterns,

contrasts, and trends that can then be interpreted within broader social and ideological frameworks.

6. Integrating Theory, Data, and Application

The three reviewed works collectively illustrate that the importance of computational discourse analysis lies in its integrative nature. Theory, data, and application are not independent components but mutually reinforcing dimensions of the field. • Cognitive theories of inference guide the design of discourse models. • Annotated corpora translate abstract discourse concepts into empirical data. • Applied computational studies demonstrate the explanatory power of these models in social contexts. This integration allows computational discourse analysis to overcome the limitations of both purely qualitative and purely quantitative approaches. It enables systematic analysis without sacrificing theoretical depth, and scalability without abandoning interpretive insight.

7. Implications and Future Directions

The growing importance of computational discourse analysis has several implications for future research. First, there is a need for hybrid models that combine inferential reasoning with data-driven learning. Second, the development of discourse resources for under-resourced languages remains a critical priority. Third, computational discourse analysis offers new opportunities for interdisciplinary research, particularly in the study of technology, politics, and media discourse. As artificial intelligence increasingly mediates communication, understanding how discourse constructs meaning, authority, and emotion becomes more urgent. Computational discourse analysis provides a framework capable of addressing this challenge.

8. Conclusion

This review has argued that computational discourse analysis represents a fundamental shift in how discourse is studied in the digital age. By integrating cognitive theories of inference, linguistically grounded data resources, and large-scale computational methods, the field offers a comprehensive approach to understanding discourse as a structured, inferential, and socially embedded

phenomenon. Rather than positioning itself in opposition to traditional discourse analysis, computational discourse analysis extends its scope and relevance. It enables scholars to examine meaning-making processes across unprecedented scales of data while remaining grounded in linguistic and theoretical insight. As such, computational discourse analysis is not merely an emerging subfield, but a necessary evolution in the study of language and society.

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